What’s in an age-depth model?

Niklas Hohmann (N.H.Hohmann@uu.nl)

Christian Zeeden

Mind The Gap

erc

Utrecht University

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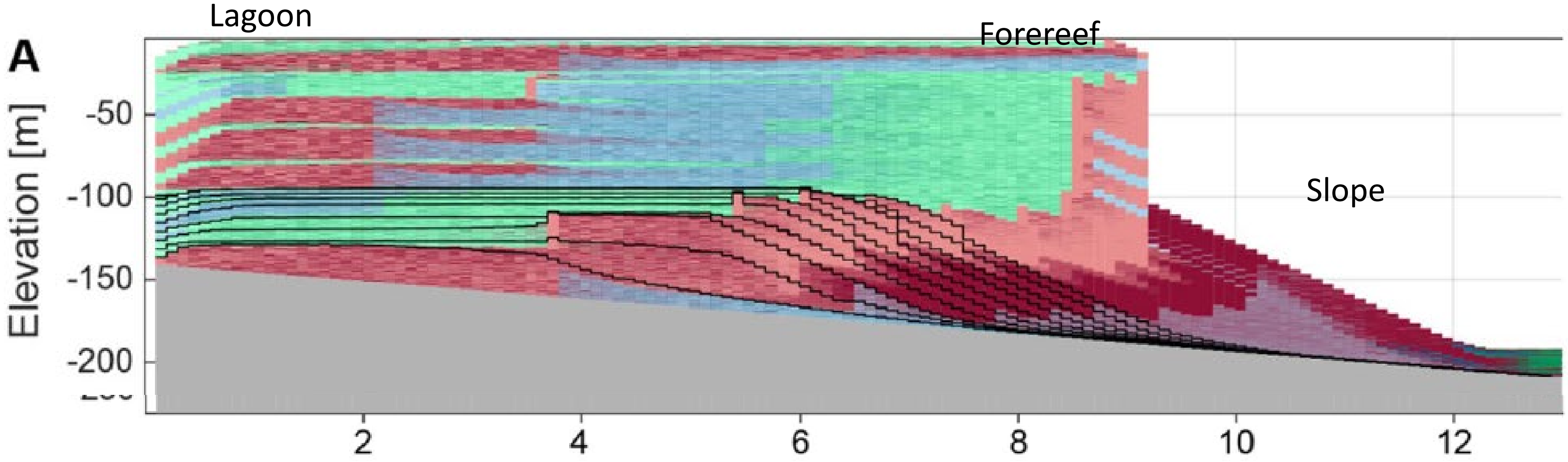
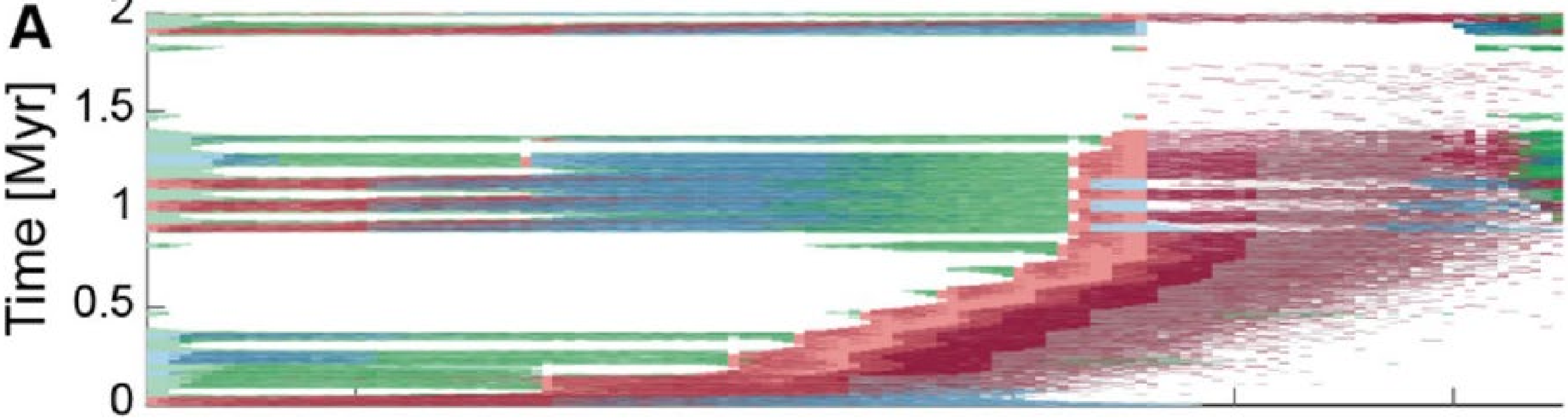
Example data: Carbonate Platform

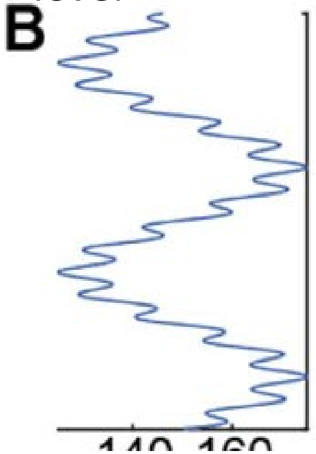
Basin transect



Chronostratigraphic (Wheeler) diagram

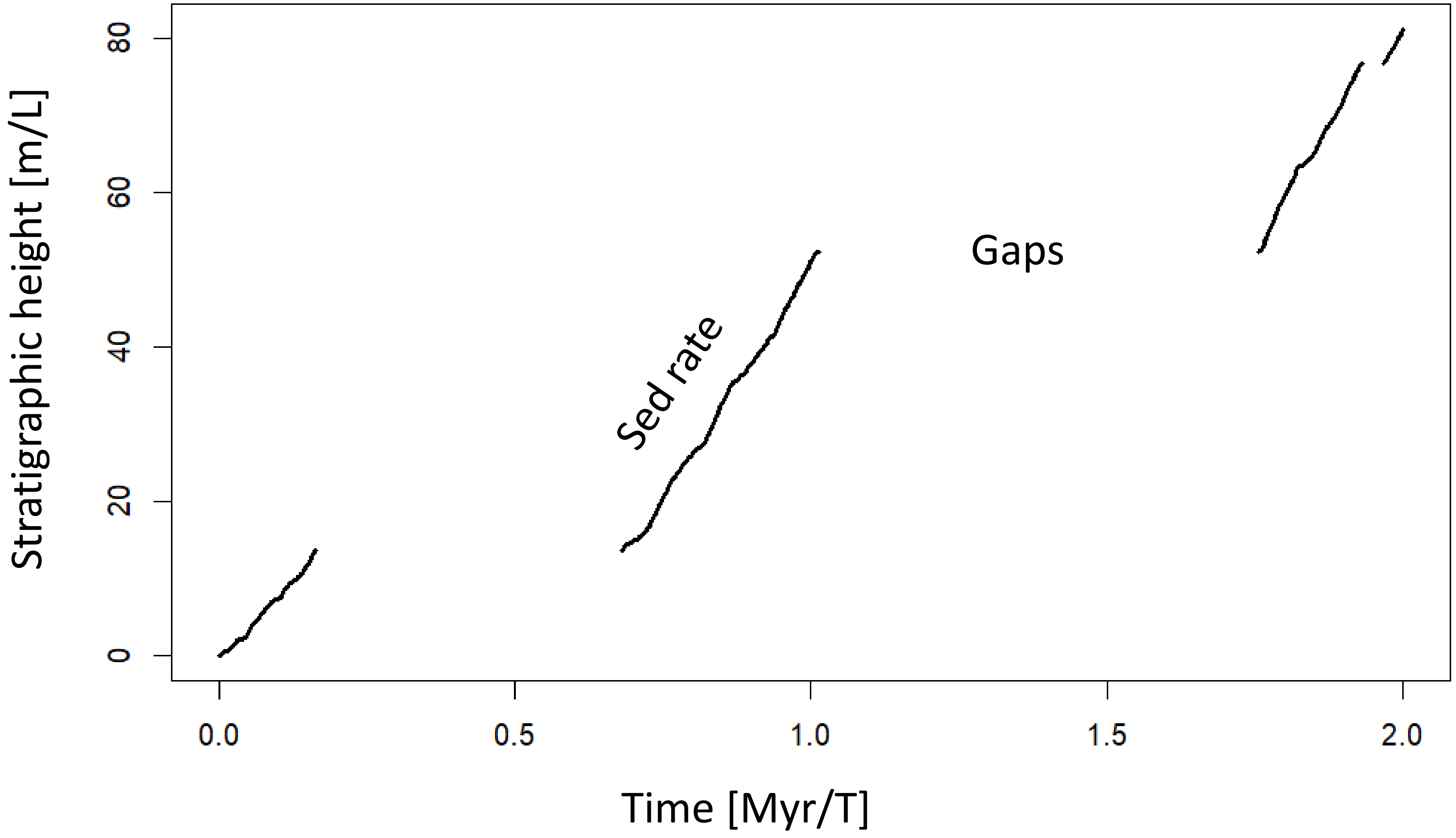


Eustatic sea level



Hohmann et al. (2024)

Age-depth model at 2 km



Hohmann et al. (2024)

ADMs are coordinate transformation between time and depth domain!

Sedimentation rates [L/T]

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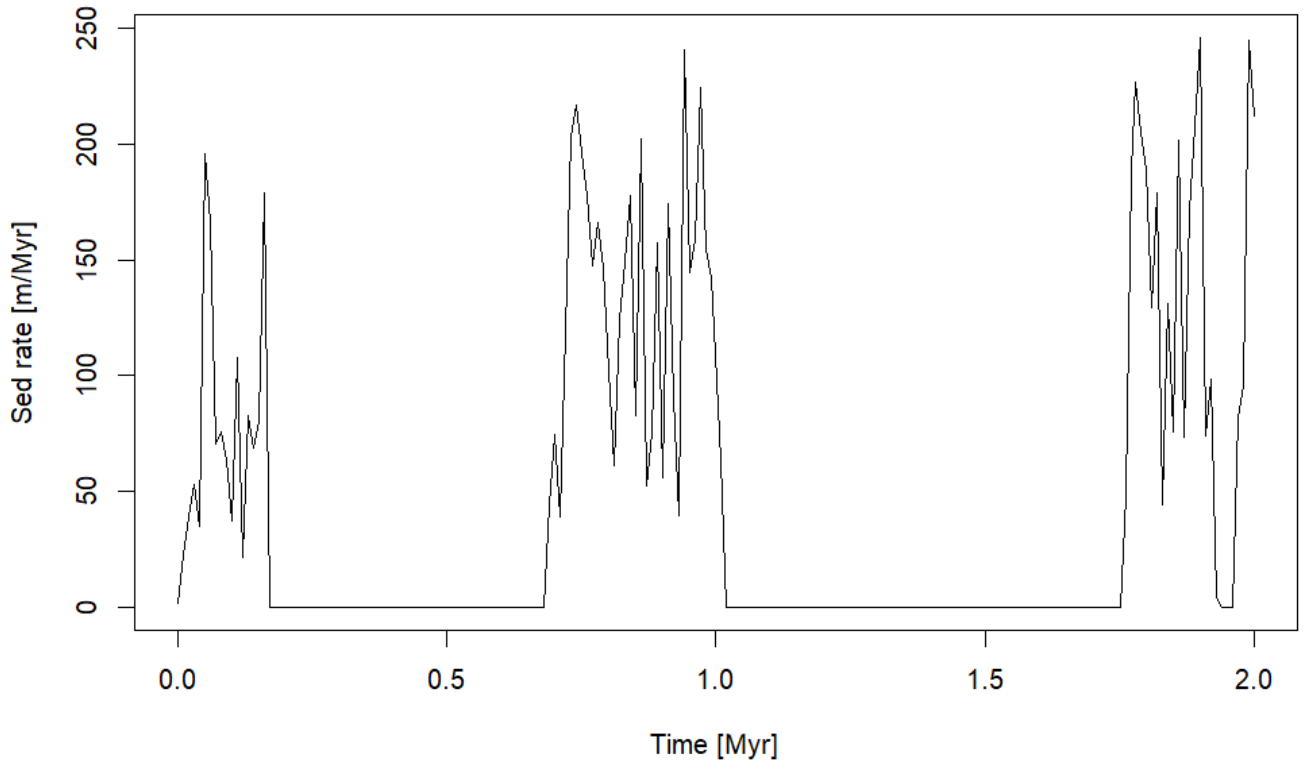
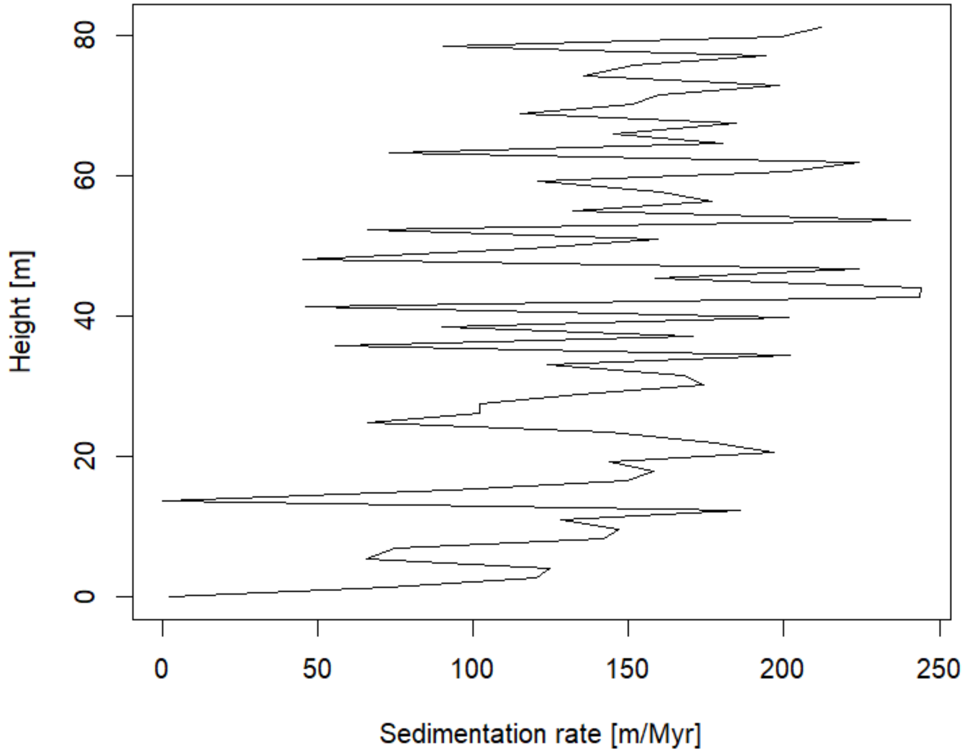
- In time domain



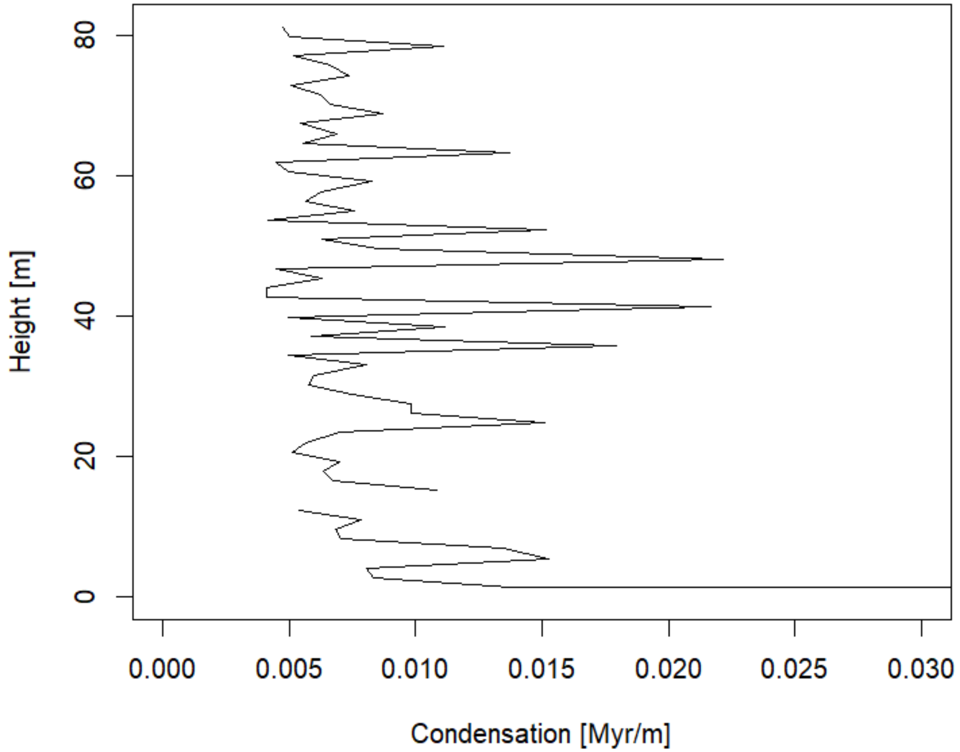
Sedimentation rates [L/T]

- In time domain
- In stratigraphic domain

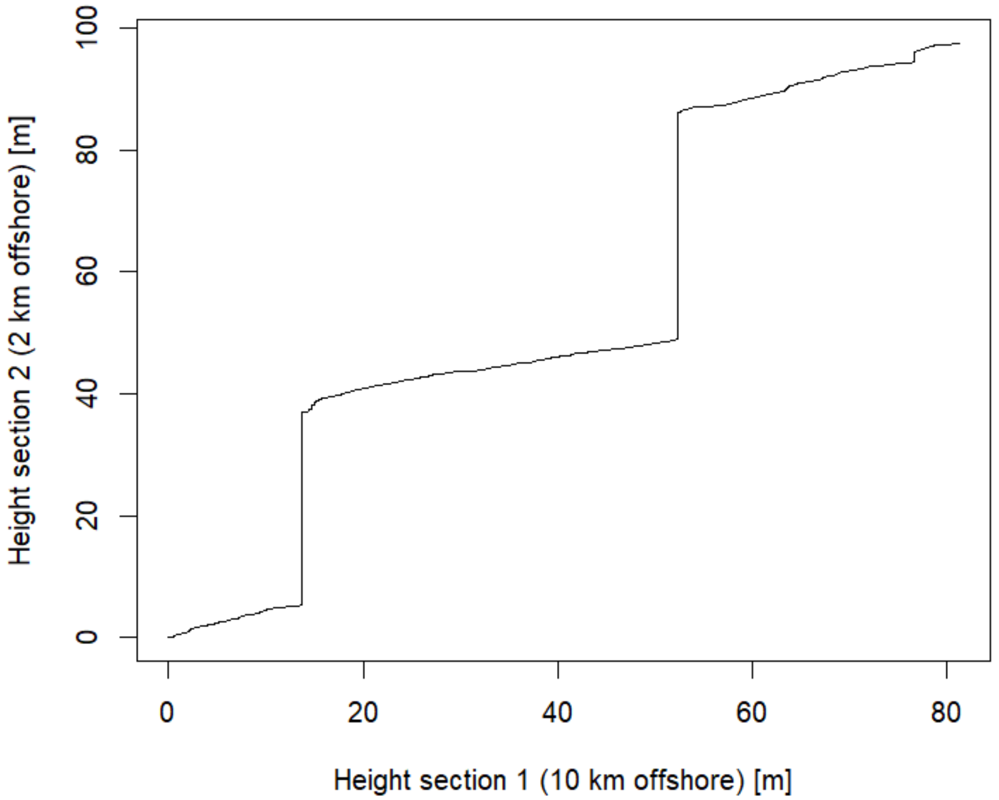


Sedimentation rates [L/T]

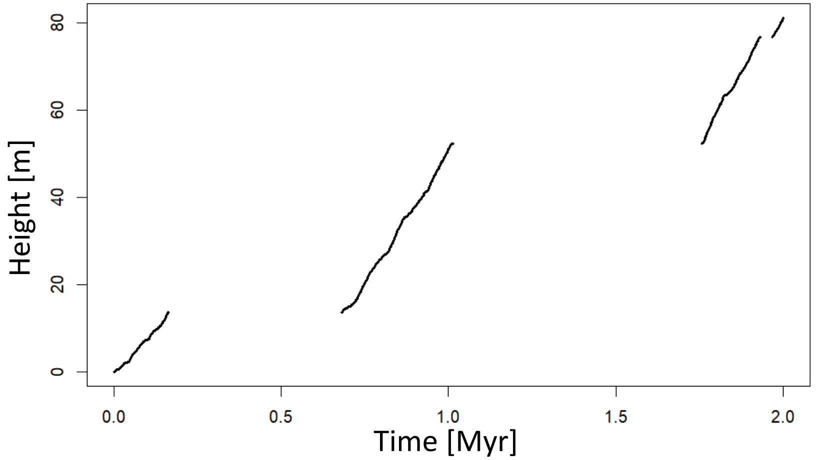
- In time domain
- In stratigraphic domain
- Better: condensation – time per height [T/L]



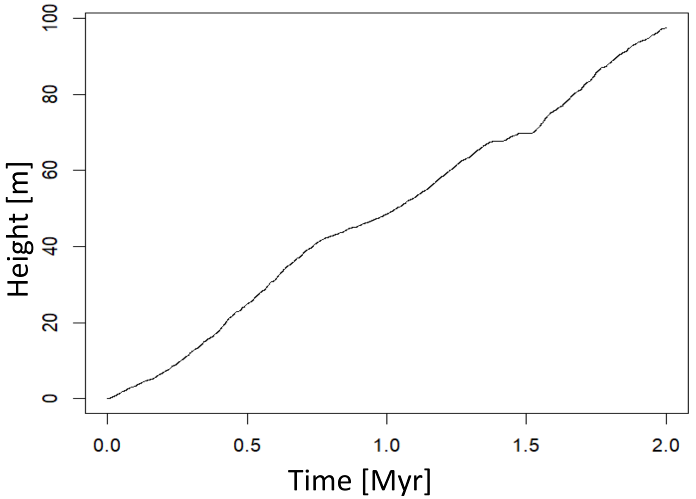
Correlation is age-depth modeling



2 km from shore



10 km from shore



Wrong ADMs, wrong inferences

- Error propagation into all age-related interpretations
- Half the sedimentation rate -> double the rate

Practical (45 mins, groups of 2)

DarwinCAT

Stratigraphic paleobiology tab

- How are different evolutionary histories preserved?
- How does this change from onshore to offshore?
- How is sea level preserved? What misinterpretations are possible?

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Shellbed condensator

- Generate a shellbed from constant shell input
- Have a peak of shell input in the time domain, but constant shell concentration in the stratigraphic domain
- Interpret sedimentation rate (both domains), age-depth model, and condensation. What happens above/below hiatal surfaces?

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